Kritika Prasai

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Summary

I am a Computer Science Ph.D. student at the University of Missouri–Columbia. My research integrates deep learning and language models with computational biology to tackle key challenges in bio-informatics and genomics. I focus on protein–protein interaction modeling, contrastive and unsupervised learning for small-sample datasets, and genomic and single-cell analyses aimed at advancing strategic antigenic variation to improve influenza vaccine effectiveness.

Education

Ph.D. student: Computer Science — Aug 2022 – Present
University of Missouri, Columbia, MO — *Advisor: Dr. Henry Wan*

- **GPA**: 4.0/4.0

Bachelor's in Computer Engineering — 2015 - 2020
Tribhuvan University, Nepal

- **GPA**: 3.5/4.0

Experience

Graduate Research Assistant — Aug 2022 – Present
University of Missouri, Columbia, MO

- **Multi-Task Protein Binding Prediction:** Developed a multi-stage cross-attention neural network for predicting SARS-CoV-2 host susceptibility through joint classification and regression of ACE2-RBD binding affinity. Integrated protein language model (ESM-2) with novel hierarchical cross-attention architecture to capture residue-level interaction patterns. Implemented advanced techniques including gated attention mechanisms outperforming sequence-similarity baselines.

- **Epistasis-Aware Modeling for Antigenic Escape:** Working on developing a structure-aware deep learning model to quantify variant-specific and epistatic effects at key influenza antigenic residues. Utilizing protein language model (ESM-2) embeddings combined with physicochemical and structural features to learn nonlinear genotype–phenotype mappings and adding a mixture-aware epistasis network to model variant proportion effects and predict immune escape dynamics.

- **Genomic Analysis for Vaccine Effectiveness:** Conducted large-scale viral sequence analyses to quantify polymorphisms and entropy-based diversity, revealing antigenic evolution patterns. Integrated genomic and single-cell RNA-seq data to characterize immune responses and inform **strategic antigenic variation** for improved influenza vaccine breadth and durability.
- **Differential Expression Analysis:** Performed comprehensive differential expression analysis on RNA seq data to identify significant gene expression changes between two influenza A and D viruses. Applied advanced statistical techniques such as generalized linear models (GLMs), quasi-likelihood F-tests, and exact tests to evaluate differential expression and used volcano plot and gene enrichment and gene-network analysis for visualization of results .

AI Engineer — Feb 2020 – July 2022
PSP Corporation, South Asia Liaison Office, Lalitpur, Nepal

- **Multi-Organ Segmentation:** Developed a two-stage deep learning Bayesian U-Net model for 3D image segmentation of organs including the spleen, liver, kidneys, and kidney cysts. Achieved Dice and Jaccard scores in the range of 92 to 96%, comparable to state-of-the-art models, using real patient data from a Japanese hospital. This model aids doctors in precisely calculating organ volumes and provides an additional layer of verification for diagnoses.

- **Pseudo-Labeling for Cost-Efficient Annotations:** Conducted a study on the application of Pseudo-Labeling techniques in medical image segmentation. This approach enabled cost-effective annotations through semi-supervised learning, reducing the need for extensive manual labeling while maintaining high accuracy in medical imaging tasks.

◦ **Lung Fissure Detection:** Explored the use of Deep Neural Networks for detecting lung fissures, a critical step in lung lobe segmentation. This ongoing project aims to improve the precision of lung segmentation, leveraging real patient data to refine and validate the model's effectiveness.

Publications

Tang, C.Y., Gao, C., **Prasai, K.**, Li, T., Dash, S., McElroy, J.A., Hang, J., Wan, X.-F. *Prediction models for COVID-19 disease outcomes.* Emerg Microbes Infect. 2024;13(1):2361791.
doi: https://doi.org/10.1080/22221751.2024.2361791 . PMID:38828796; PMCID:PMC11182058.

Prasai, K., Yang, Z., Guan, M., Li, T., Ware, D., Hang, J., Wan, X.-F. *Intrahost HA polymorphisms and culture adaptation shape antigenic profiles of H3N2 influenza viruses.* J Virol. 2026 Jan 7:e0177525. doi: 10.1128/jvi.01775-25 . Epub ahead of print. PMID:41498543; PMCID:PMC12831994. *Supported by: NIH & NIAID [R01AI147640]*

Wan, X.-F., Guan, M., Balamalaliyage, P., Chen, H., **Prasai, K.**, et al. *Epitope-spanning antigenic variation reprograms immunodominance and broadens immunity in sequential influenza vaccination.* Nature Communications. doi: 10.1038/s41467-026-70202-y Supported by: NIH [R01AI152521 & R01AI147640]

Espada, C., Ye, K., Long, Y., **Prasai, K.**, DeLiberto, T., Heale, J., Wiese, R., Yang, Q., Zhou, M., Streich, S., Tao, Y., Chandler, J. *Species- and variant-specific ACE2 compatibility shapes SARS-CoV-2 spillover potential in North American cervids.* https://doi.org/10.1038/s41467-026-71623-5 Nature Communications supported by U.S. Department of Agriculture Animal and Plant Health Inspection Service .

Prasai, K., Chandler, J.C., Espada, C., Wiese, R., Long, Y., Streich, S.P., Heale, J., Roberts, N., Tao, Y.J., DeLiberto, T.J., Wan, X.-F. *Interaction-aware multitask deep learning reveals cross-species receptor compatibility landscapes across sarbecoviruses.* Nature Communications biology (under review).

Programming Skills

→ **Languages:** Python, C/C++, R, JavaScript

→ **Infrastructure & Systems Tools:** Git, MySQL, Apache Spark, Docker, CUDA

→ **Machine Learning & Deep Learning Frameworks:** PyTorch, TensorFlow, Keras, Scikit-learn, Fastai

→ **Large Language Models (LLMs):** Hugging Face Transformers, OpenAI API, LangChain, LLaMA, GPT models, BERT; **Protein LLMs:** ESM, ProtBERT

→ **Bioinformatics:** scRNA-seq , whole-genome sequencing, multi-omics data integration, sequence assembly pipelines, SNV and genomic variant calling analysis

→ **Data Science & Analytics Tools:** NumPy, Pandas, Matplotlib, Seaborn, Excel

Achievements and Accolades

- **2026 Rising Stars in Computational & Data Sciences Workshop:** Selected for a research talk at the Santa Fe Institute workshop recognizing emerging PhD researchers in computational and data sciences.
- **2026 Electrical Engineering and Computer Science Outstanding PhD Student Award:** Recognized by the EECS department at the University of Missouri for outstanding research, academic excellence, and contributions to the graduate student community.
- **Interdisciplinary Case Competition 2024 Winner:** Organized by the Graduate Professional Council (University of Missouri).
- **American Society of Virology (ASV 2024):** Oral presentation on Ph.D. research focused on differential expression analysis of innate immune responses.
- **Epidemics 2025:** Poster presentation on intrahost polymorphisms and culture adaptation shaping antigenic profiles of H3N2 influenza viruses.

Student Leadership & Services

- **President, Electrical Engineering and Computer Science GSA (2025–2026):**
  - → Led graduate student engagement initiatives, including organizing the **Mizzou Tiger Internship Seminar** to provide a platform for students to share their internship experience with their peers.
  - → Founded and organized the inaugural **ShowMe Course Project Showcase**, enabling students to present semester-long projects to faculty judges and compete for awards.
  - → Coordinated a college-wide hackathon focused on the ARC-AGI dataset, fostering interdisciplinary collaboration and innovation.
  - → Organized a **Career Pathways Panel** by inviting Mizzou alumni from industry, academia, and national laboratories to share their career journeys, application strategies, and challenges, helping students better prepare for diverse career paths.
  - → Served as an Engineering judge (via Ideal-Logic) for high school student research projects organized by the Mizzou College of Engineering, evaluating and mentoring emerging student researchers.
- **Department Representative, Graduate Professional Council (GPC):** Represented EECS on the Graduate Professional Council; contributed to the Student Affairs Subcommittee advocating for graduate student mental health and well-being through campus-wide policies and initiatives.
- **Vice-President, University of Missouri Nepalese Student Association (MUNSA) (2025–2026):**
  - → Led organization of new student welcome and orientation programs, mentoring incoming Nepalese students to support their academic and social integration into the university community.
  - → Co-ordinated cultural and community events, including a women's empowerment showcase highlighting Nepalese traditions and promoting inclusivity and representation.